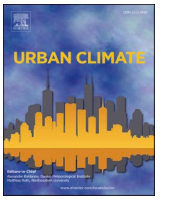
Urban Climate 58 (2024) 102138



Contents lists available at ScienceDirect

Urban Climate

journal homepage: www.elsevier.com/locate/uclim





Integrating thermal perception and public space use – an experimental outdoor comfort study in cold winter-hot summer zone: Beijing, China

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ARTICLE INFO

Keywords:
Urban public space
Outdoor thermal comfort
Urban design
Questionnaire survey
Microclimate
Physiological equivalent temperature (PET)

ABSTRACT

This study aims to unravel the seasonal differences in outdoor thermal comfort (OTC) in a cold winter-hot summer region, as well as the comprehensive impact of meteorological and human factors, along with people's activity patterns in public spaces, on OTC. Questionnaire surveys, microclimate measurements, and photo recording were conducted over 21 daytime sessions in seven public spaces across winter and summer. The results indicated a significant seasonal OTC variance in Beijing. The neutral PET and neutral PET range were 21.9°C and 15.2–28.6°C for summer, and 14.3°C and 6.1–22.5°C for winter. Globe temperature primarily impacted OTC during both seasons, while wind speed was the only additional influential factor in winter. In Beijing, summer cooling is more crucial than winter insulation; however, winter wind protection is prioritized over summer ventilation. Some human factors, including age, activity level, visiting purpose, short-term thermal experience, and environmental stimuli, also influence OTC to some extent. Furthermore, this study identified significant variations in the temporal distribution and age demographics of attendees in various types of public spaces across different seasons, which considerably influence OTC. Our findings provide insights into the relationship between outdoor thermal comfort and public space use, which informs climate adaptive urban design.

1. Introduction

Thermal comfort significantly affects people's willingness to engage in outdoor activities and the quality of urban spaces. Understanding urban outdoor thermal comfort (OTC) conditions is necessary for building a comfortable, healthy, and livable city. The OTC comprises two primary components: the objective microclimate condition and subjective human thermal perception (Nikolopoulou and Steemers, 2003). Both are affected by urban built environments (Nikolopoulou et al., 2001).

On the one hand, many studies have been conducted on outdoor microclimates and how they affect people's thermal comfort. Scholars have explored the contribution of various meteorological parameters on OTC, such as air temperature, solar radiation, wind and humidity (Lin et al., 2011; Ng and Cheng, 2012; Li et al., 2016; Liu et al., 2022; Tian et al., 2022). Also, the influence of built environment, such as urban geometry, vegetation, reflective surfaces, and waterbodies, on outdoor microclimate were investigated (Abd Elraouf et al., 2022; Taleghani, 2018; Lai et al., 2019; Mohammad et al., 2021). On the other hand, even under similar climatic conditions, differences in human body's thermal perception can cause significant discrepancies in thermal comfort (Nikolopoulou and

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https://doi.org/10.1016/j.uclim.2024.102138
Received 31 May 2023; Received in revised form 7 September 2024; Accepted 10 September 2024
Available online 16 September 2024
2212-0955/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

OTC	outdoor thermal comfort
PET	physiological equivalent temperature (°C)
NPET	neutral PET (°C)
NPERT	neutral PET range (°C)
TSV	thermal sensation vote
MTSV	mean TSV
TCV	thermal comfort vote
TAR	thermal acceptance range
T_g	globe temperature
T_a	air temperature
RH	relative humidity
V_a	wind speed

Steemers, 2003). Nikolopoulou (2001) first presented the thermal adaptation theory to explain this discrepancy in the OTC. Residents adapt to the climate in which they live for long periods (Zhang et al., 2022). In addition to long-term thermal history, other human physiological and psychological characteristics, such as age (Yung et al., 2019; Larriva et al., 2020), gender (Jin et al., 2019), short-term thermal experience (Ng and Cheng, 2012; Yung et al., 2019), activity intensity (Huang et al., 2016), autonomy of activity (Zabetian and Kheyroddin, 2019), length of stay in outdoor place (Chen et al., 2015), environmental stimuli (Lam et al., 2020; Lau and Choi, 2021), thermal adaptive behaviors (Huang et al., 2019; Kim and Brown, 2022), even social factors, such as income or educational level (Tian et al., 2022) also effect OTC.

In different regions of the world, climatic conditions vary greatly, and even within the same region, there can be significant seasonal climate differences. Studies on thermal comfort in different climatic zones sometimes produce different, or even contradictory, conclusions. Many studies have proved that people have distinctive range of comfortable temperature (Salata et al., 2016; Hadianpour et al., 2018; Cohen et al., 2019; Yin et al., 2021; Liu et al., 2023) and thermal preference (Johansson et al., 2017; Tian et al., 2022) under various regions and seasons, which is caused by long-term thermal adaptation. The impact of various human factors on thermal comfort also varies in the conclusions of different studies (Lai et al., 2020a). Also, the influential meteorological parameters on OTC may vary in distinct climate conditions. For example, some studies have indicated that relative humidity is not correlated with OTC (Li et al., 2016; Zhu et al., 2022; Kumar and Sharma, 2022), whereas others have contrasting conclusions (Chow et al., 2016; Yuan et al., 2023). Whether wind speed can significantly affect thermal comfort has led to different conclusions in studies across various places, with some findings suggesting that wind speed is more influential in cold climates (Liu et al., 2016; Liu et al., 2022; Xie et al., 2022; Tian et al., 2022).

Therefore, research conclusions of this field are typically distinctive in different regions, which demonstrates the importance of targeted OTC research based on local climate regions. However, most existing studies have focused on hot climate regions. Few have been conducted on China's cold winter-hot summer climate to explore the OTC seasonal differences, and comprehensively investigated the thermal preference, influential climatic and human parameters on OTC during distinct seasons within this climatic region (Table 1).

Even in same climate zone, people may have different activity pattern (Vidal et al., 2022) and different thermal comfort condition (He et al., 2023) in variance urban outdoor spaces. This may not only be caused by climatic condition, but the surrounding urban land use and facilities as well. Li et al. (2016) surveyed the activity times of people in residential communities in a subtropical climate and found that people tend to use outdoor spaces closer to midday in winter than in summer. Lai et al. (2020b) investigated the temporal

Table 1
Recent thermal comfort researches in comparative climate region

Authors	Year	Region	Climatic characteristics	Season	Content
Chen et al.	2018	Harbin	Dwa, colder than Beijing	All year	a,b,c
Jin et al.	2019	Harbin	Dwa, colder than Beijing	All year	a,c
Yin et al.	2021	Harbin	Dwa, colder than Beijing	All year	a,b
Xu et al.	2019	Xi'an	Cwa, warmer than Beijing	Summer and winter	a,b
Tian et al.	2022	Xi'an	Cwa, warmer than Beijing	All year	a,b,c
Yuan et al.	2023	Xi'an	Cwa, warmer than Beijing	All year	b,c
Liu et al.	2023	Xi'an	Cwa, warmer than Beijing	All year	a
Ma et al.	2021	Mizhi	Dwa, a rural area but not city	winter	a
Lai et al.	2014	Tianjin	Dwa, similar to Beijing	All year	a
Lai et al.	2020b	Tianjin	Dwa, similar to Beijing	All year	c
Liu et al.	2022	Tianjin	Dwa, similar to Beijing	Autumn and winter	b
An et al.	2021	Beijing, Xi'an, Hami	Dwa (Beijing)	Winter	a,b
Xie et al.	2022	Beijing	Dwa	Summer	b,c

a. thermal comfortable range and climate adaptation; b. meteorological factors impact OTC; c. human factors impact OTC

distribution and activities of people in an urban park in different seasons. They also found that people tended to use the park closer to midday during the winter. Yuan et al. (2023) compared people's different activities patterns between urban and rural public spaces and their thermal perception variance. However, studies on the relationship between different public spaces and OTC services are relatively few and require further investigation. In Chinese cities, residences and offices are two typical urban land uses. However, there is still a lack of investigation into whether people in these two types of urban areas have different activity patterns and distinct thermal comfort conditions.

Thus, this study presented two research questions:

1) to clarify the seasonal OTC difference in Beijing, a city with cold winters and hot summers, including an analysis of the thermal comfort range, thermal adaptation, and the climatic and human factors influencing OTC during different seasons.

2) to investigate the differences in activity patterns, the presence of people, and thermal comfort between different types of urban public spaces.

This study initially collected data on microclimate conditions and human thermal perception in Beijing's urban public spaces through a questionnaire survey and on-site microclimate measurements. Additionally, it analyzed the daily temporal distribution of people in different types of public spaces using on-site observations. Following this, the study explored the conditions of OTC and their seasonal variations through detailed data analysis. It also investigated the meteorological and human factors affecting OTC under the local climatic conditions. The findings presented in this work aim to enrich the body of research on OTC in the cold winter and hot summer regions of China, and provide theoretical insights for OTC studies, as well as practical recommendations for urban design practices.

2. Methodology

2.1. Study area and climate condition

This study was conducted in Beijing, a metropolitan located in northern China (39.9°N, 116.4°E) with over 20 million population. Beijing belongs to Dwa (humid continental climate) zone in Köppen-Geiger climate classification. The climate is characterized by hot humid summer and cold dry winter, which may cause substantial discrepancy on human thermal perception in two seasons. Fig. 1 indicates Beijing's average meteorological data from 2003-2022, which was obtained from the website of National Meteorological

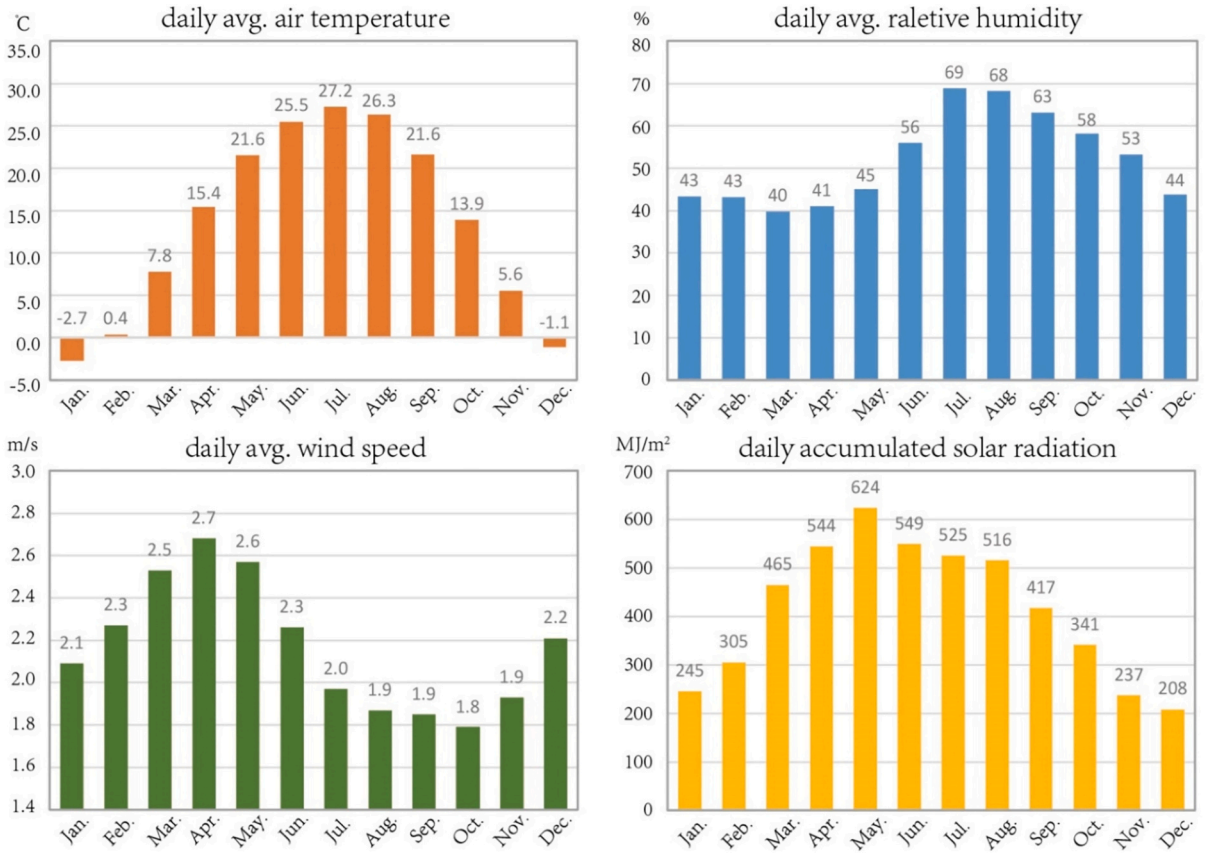


Fig. 1. The change of four meteorological factors during a year in Beijing.

Information Center (http://data.cma.cn/). The air temperature is the highest on July, followed by August and June, and the lowest on January, followed by December and February. The climate of autumn and spring are relatively moderate and comfortable. Thus, the on-site investigation campaign of this study was conducted during July and August in summer, and during December, January and February in winter.

2.2. Field study

On-site investigations, including microclimate measurement, questionnaire survey, and photo recording, were conducted in this study to obtain OTC data.

2.2.1. Selection of public space

Seven public spaces in Beijing were selected for the field investigation. The nearest location was approximately 4 km from the city center, whereas the farthest distance was 15 km. To investigate the differences in the physiological, psychological, and activity characteristics of users in different types of public spaces, the seven sites were divided into three types according to their surrounding urban land use: residential area (sites 1/2/3), office area (sites 4/5/6), and mixed area (site 7), as shown in Fig. 2a. The reasons for selecting these sites are as follows. Firstly, all sites were densely populated and easily accessible, ensuring an adequate number of questionnaires and the reliability of time-varying population data. Secondly, the sites were located in different areas of Beijing, ensuring diversity in the characteristics of the site users. Lastly, all sites had a certain scale, which ensured a variety of activities for the users.

2.2.2. On-site microclimate measurements

Two portable microclimate stations were used in this study to collect the weather data in each public space during on-site investigations (Fig. 2b). A Misol hp2550 weather station was used to capture air temperature (T_a), relative humidity (RH), and wind speed (V_a) data. A Jianda Renke 50mm diameter globe temperature sensor was used to obtain globe temperature (T_g). All sensors were fixed at approximately 1.5 m height above the ground by tripods. All meteorological parameters were recorded in 5min interval. The location of sensors at each public space is illustrated in Table 2. Beside air temperature and solar radiation, ground reflectance and wind speed are significant factors influencing OTC. Therefore, all measurement points were located in well-ventilated areas, with the gray hard ground pavement.

2.2.3. Questionnaire survey

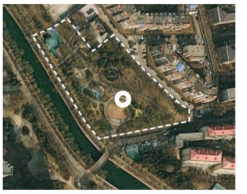
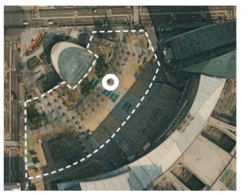
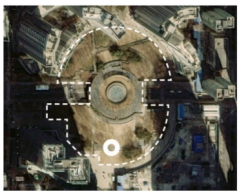
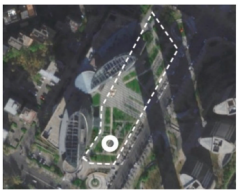
The objective of the questionnaire survey was to obtain information on citizens' body characteristics, activities, and thermal perceptions. Citizens participating in free activities at the study sites were asked to complete questionnaires, whereas meteorological parameters were measured. Each questionnaire was recorded with the time of completion to facilitate matching with the results of microclimate measurements. All the respondents were located within 10 m of the measuring instrument and were guaranteed the same sunlight conditions as the sensors (Fig. 2c). Additionally, when the solar radiation conditions at the measurement points changed (due to shadows cast by buildings or trees), the questionnaire survey was paused for 10 minutes. It was resumed once the measurement values stabilized.

The questionnaire was designed according to previous studies (Ng and Cheng, 2012; Chen et al., 2015; Shooshtarian and Ridley, 2016; Lai et al., 2020b; Tian et al., 2022) and divided into two parts. The first part focused on human factors that may influence thermal perception, including the respondents' physiological conditions, such as gender, age, long-term thermal experience (length of residence in Beijing), short-term thermal experience (whether in an air-conditioned place in the past 15 min), and type of activity. It also examined psychological factors, such as the purpose of activity, the length of stay in that place, whether they stayed alone or were



Fig. 2. Locations of seven selected sites (a), microclimate sensors (b) and questionnaire survey process (c)

Table 2
Description of 7 selected public spaces

Site No.	Type	Description	Satellite map	Photograph
1	Residential	A small urban park, surrounded by mid-rise residential buildings		
2	Residential	A small urban park along the river, with mid-rise residential buildings nearby		
3	Residential	An urban square in the corner of a cross-road, some high-rise buildings nearby		
4	Office	A curve-shaped square beside a high-rise office tower and its podium building		
5	Office	A small, circular-shaped urban park surrounded by high-rise office buildings		
6	Office	A strip-shaped urban square flanked by high-rise office buildings, featuring good greenery.		
7	Mixed	A square surrounded by high-rise office towers and residential buildings, with well-maintained greenery.		

accompanied, environmental stimuli (the scenery of the place) and overall satisfaction. The second part explored the respondents' thermal perceptions, including thermal sensation vote (TSV), thermal comfort vote (TCV), discomfort on each meteorological factor, and thermal preference on each meteorological factor. The TSV was adapted from ASHRAE's 7-point scale (very cold = -3, cold = -2, slightly cold = -1, neutral = 0, slightly hot = 1, hot = 2, very hot = 3). The TCV, thermal discomfort and thermal preference were recorded on a 3-point scale. Appendix 1 presents the detailed design of the questionnaire.

It should be noted that in Question 4, the length of residence was classified into two groups in the data analysis: less than one year and more than one year. In Question 6, the type of activity was classified into three groups according to intensity (Huang et al., 2016). Sitting, standing, and looking after children were classified as low-intensity activities; walking slowly was classified as a mid-intensity activity; fast walking, and doing exercise were classified as high-intensity activities. In Question 8, visiting purposes were classified into three groups according to the autonomy of the activities (Lai et al., 2020a). Doing group exercises, meeting friends, and taking care of children were classified as social activities, which had a high degree of autonomy; doing exercise alone and rest/recreation were classified as spontaneous activities, which had a moderate degree of autonomy; walking by, waiting for someone, and working were classified as necessary activities, which had the lowest degree of autonomy.

2.2.4. Human observation in public spaces

In addition to microclimate measurements and questionnaire surveys, changes in the number of users of public spaces over time were recorded to study the temporal distribution of people in different seasons and types of public spaces. We chose to take photos of all users of the public space three times every hour (e.g., 9:10, 9:30, and 9:50). The total number of people in the photos was accounted. Then, the number of people at the three time points was averaged to calculate the average number of public space users per hour. Additionally, this study examined the age distribution characteristics of users in different types of public spaces. By observing photographs, we recorded the number of elderly individuals in each photo and calculated the total number of elderly individuals present in the public space throughout the day. Then the proportion of elderly individuals relative to the total number of users was determined. It is important to note that, since it was not possible to accurately determine age through observation, the proportion of elderly individuals obtained by this method was compared with the proportion of respondents over 55 years old from the questionnaire to ensure the validity of the conclusions.

2.2.5. Field survey schedule

Twenty-one field surveys were conducted during winter (11 times) and summer (10 times). The dates, times and measured weather conditions of field survey is listed in Table 3. To study the temporal distribution of users in public spaces at different times, one weekday and one weekend were selected for investigating residential and mixed-use public spaces. As offices are used less frequently on weekends, on-site investigations were conducted only on weekdays in each office public space. None of the investigations were conducted under rainy or snowy days.

2.3. Thermal sensation index and calculation

This study applied the physiological equivalent temperature (PET) as the thermal sensation index to calculate the objective human thermal perception. According to Deng et al. (2023), 30.2% of the OTC studies chose PET as the primary thermal sensation index. Therefore, using PET as an indicator of OTC contributes to the comparison with similar studies. The applicability of PET has been

Table 3
Date, times and weather conditions of field survey

Season	Survey date	Time	Site No.	Sky condition	Max. T_a (°C)	Avg. V_a (m/s)
Summer	Jul. 10, 2022	09:00-19:00	residential	Cloudy	28.7	0.43
	Jul. 13, 2022	09:00-19:00	residential	Sunny	35.6	0.82
	Jul.15, 2022	09:00-19:00	mixed	Sunny	38.4	0.47
	Jul.17, 2022	09:00-19:00	mixed	Sunny	37.2	0.64
	Jul.19, 2022	09:00-19:00	residential	Partly cloudy	35.0	0.20
	Jul. 21, 2022	09:00-19:00	office	Sunny	33.2	0.67
	Jul. 29, 2022	09:00-19:00	office	Partly cloudy	31.2	1.03
	Jul. 31, 2022	09:00-19:00	residential	Partly cloudy	34.1	0.29
	Aug. 2, 2022	09:00-19:00	residential	Sunny	35.8	0.28
	Aug. 17, 2022	09:00-19:00	office	Partly cloudy	31.1	0.31
Winter	Dec. 23, 2022	09:00-17:00	residential	Sunny	2.9	0.48
	Dec. 25, 2022	09:00-17:00	residential	Partly cloudy	1.2	0.15
	Dec. 29, 2022	09:00-17:00	residential	Partly cloudy	3.8	0.65
	Dec. 31, 2022	09:00-17:00	residential	Sunny	8.0	0.37
	Jan. 8, 2023	09:00-17:00	residential	Sunny	12.4	0.82
	Jan. 11, 2023	09:00-17:00	residential	Cloudy	2.7	0.23
	Feb. 1, 2023	09:00-18:00	office	Partly cloudy	3.6	0.45
	Feb. 3, 2023	09:00-18:00	office	Sunny	5.0	0.27
	Feb. 5, 2023	09:00-18:00	mixed	Partly cloudy	6.3	0.54
	Feb. 7, 2023	09:00-18:00	office	Sunny	6.4	0.39
	Feb. 14, 2023	09:00-18:00	mixed	Cloudy	1.7	0.22

approved by past studies in various climatic regions (Potchter et al., 2018; Cohen et al., 2019; Canan et al., 2020; Kumar and Sharma, 2022), and similar climatic regions of Beijing (Lai et al., 2017; Tian et al., 2022).

Rayman (version 1.2) software was used to calculate the PET index. This software was developed by Prof. Matzarakis of Albert–Ludwigs-University of Freiburg (Matzarakis et al., 2007) and has been widely applied in OTC studies (Potchter et al., 2018; Larriva and Higueras, 2020; Labdaoui et al., 2021). Tmrt (mean radiant temperature), an important input parameter to calculate PET for the Rayman software, was obtained using the following formula (1) according to the ISO 7726 standard (2002):

$$T_{mrt} = \left[(T_g + 273)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{0.25} - 273 \quad (1)$$

where T_g is globe temperature (°C), T_a is air temperature (°C), V_a is wind speed (m/s), D is globe diameter (D=0.05 m), and Ɛ is the emissivity (Ɛ=0.95).

3. Results and data analysis

During the field survey, 986 questionnaires were given out during the survey campaign and 975 valid questionnaires were collected (494 in summer and 481 in winter). A total number of 21379 people were recorded (10871 in summer and 10508 in winter). During microclimate measurement, T_a was recorded between 22.2°C to 38.3°C in summer and -4.3°C to 12.4°C in winter, RH was recorded between 22% to 87% in summer and 16% to 75% in winter, V_a was recorded between 0m/s to 2.8m/s in summer and 0m/s to 2.1m/s in winter, T_g was recorded between 23.8°C to 48.5°C in summer and -2.6°C to 19.2°C in winter, PET was calculated between 22.8°C to 55.2°C in summer and -4.1°C to 20.2°C in winter.

3.1. Overall thermal perception in two seasons

Table 4 shows the proportion of each TSV scale during the two seasons, based on the questionnaire survey. During summer, 33% of respondents chose “neutral”, while during winter, 46.6% chose this option. 36.8% of respondents chose “slightly hot” during summer, while 41.2% chose “slightly cold” during winter. Meanwhile, 29.7% of respondents chose “hot” and “very hot” during summer, while only 8.5% chose “cold” and “very cold” during winter. The average TSV was 1.06 for summer with the standard deviation of 0.97. In contrast, the average TSV was -0.55 for winter, with the standard deviation of 0.73. Therefore, it can be concluded that there are considerable differences in thermal perception between summer and winter in Beijing. People are generally more tolerant to cold weather during winter than to hot weather during summer. This hypothesis was also supported by the comparison of TCV between the two seasons (Table 5). During summer, 34% of respondents chose “comfortable”, which was 12.4% less than during winter. Meanwhile, 8.1% of respondents chose “unacceptable” during summer, and only 2.7% of respondents chose this option during winter.

3.2. Locating NPET, NPETR and TAR

The relationship between the measured PET and mean TSV (MTSV) was examined. MTSV was calculated at an interval of 1°C PET bin separately for summer and winter, and the regression model were calculated and shown in Fig. 3. The neutral temperature refers to the temperature at which people feel neither cold nor hot (ASHRAE, 2017). According to past studies (Cheung and Jim, 2017; Tian et al., 2022; Yuan et al., 2023), neutral PET (NPET) is defined as MTSV=0, and neutral PET range (NPETR) is defined as MTSV between -0.5 and 0.5. People have the most comfortable thermal perceptions in this range. By the regression model that was developed in this study, the NPET of residents in Beijing were 21.9°C for summer and 14.3°C for winter, while the NPETR were 15.2–28.6°C for summer and 6.1–22.5°C for winter. Thus, NPET has substantial differences for the two seasons in Beijing, which reflects the discrepancy in thermal adaptation and thermal expectations of residents in Beijing, a cold-winter and hot-summer climate region. Same method was applied to obtain the NPET for public spaces near residential area and office area. The NPETs were 18.7°C (R^2=0.227, Sig.=0.00) and 26.8°C (R^2=0.530, Sig.=0.01) in summer, and 13.1°C (R^2=0.645, Sig.=0.00) and 15.3°C (R^2=0.392, Sig.=0.00) in winter for residential and office area respectively.

ASHRAE 55 (2017) defines TAR as the temperature range that at least 80% or 90% of respondents can accept. In our study, the temperature range that was acceptable to 90% of the respondents was obtained from the TCV and chosen as the TAR. The thermally unacceptable rate for every 1°C PET bin was calculated, and the TAR was obtained using a linear regression model for summer and

Table 4
Proportion of each TSV scales

Options	Proportion (summer)	Proportion (winter)
Very cold (-3)	-	1.0%
Cold (-2)	-	7.5%
Slightly cold (-1)	0.4%	41.2%
Neutral (0)	33.0%	46.4%
Slightly hot (1)	36.8%	4.0%
Hot (2)	19.2%	-
Very hot (3)	10.5%	-

Table 5
Proportion of each scale of TCV

Options	Proportion (summer)	Proportion (winter)
Comfortable (1)	34.0%	46.4%
Acceptable (2)	57.9%	50.9%
Uncomfortable (3)	8.1%	2.7%

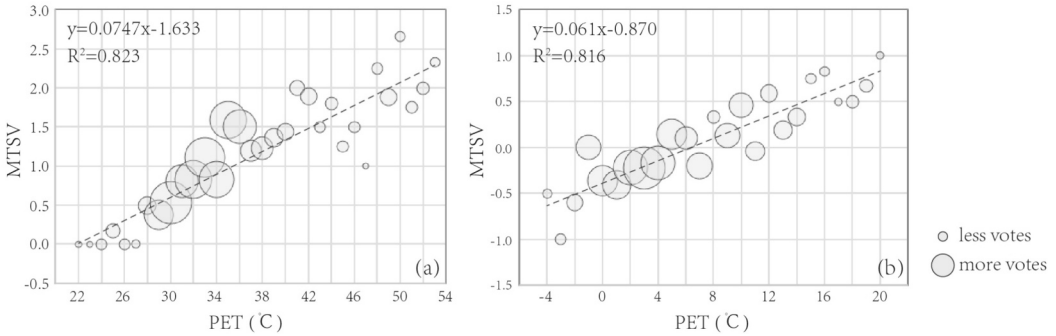


Fig. 3. Regression model of MTSV and PET for summer (a) and winter (b)

winter. The TAR of the Beijing area was 35.7°C for summer (Fig. 4). However, winter TAR was not obtained because the R^2 of the linear regression model was < 0.1. The reason may be that the winter in that year was mild; the lowest PET was -4°C. Hence, not many people felt uncomfortable.

3.3. *Meteorological factors that affect TSV*

The dominant influence of climatic factors on OTC, such as T_g, T_a, RH, and V_a, has been proven. However, the perceptions of each microclimatic factor vary under different climatic conditions (Lai et al., 2014; Deng et al., 2023). Therefore, this study obtained the weights of various microclimatic factors affecting the thermal perceptions of residents for Beijing's winter and summer. An ordinal logistic regression model was used to analyze the influence of each climatic factor on the TSV. Table 6 indicates the correlation between the climatic factors and TSV for two seasons. During summer, T_g, T_a, and RH were positively correlated with TSV, whereas V_a was negatively correlated with TSV. During winter, T_g was positively correlated with TSV, whereas V_a was negatively correlated with TSV. However, T_a and RH did not significantly influence TSV. Stepwise logistic regression was conducted to determine the weight of influence of each factor. The sequence of influential weights on TSV was T_g>RH>T_a>V_a and Tg>V_a during summer and winter, respectively.

Fig. 5 shows the percentages of thermal discomfort and preference votes during the two seasons. During winter (Fig. 5a), 20% of the respondents felt that sunlight was too weak, 26% thought that high wind speed caused discomfort, and 24% felt disturbed by low relative humidity. During summer (Fig. 5c), 43% of the respondents felt the sunlight was too strong, only 10% felt the wind was too weak, and 33% thought the relative humidity was too high to be comfortable. The thermal preference vote exhibited a regularity similar to that of the thermal discomfort vote. During winter (Fig. 5b), 39%, 43%, and 36% of the respondents wished for T_a, sunlight,

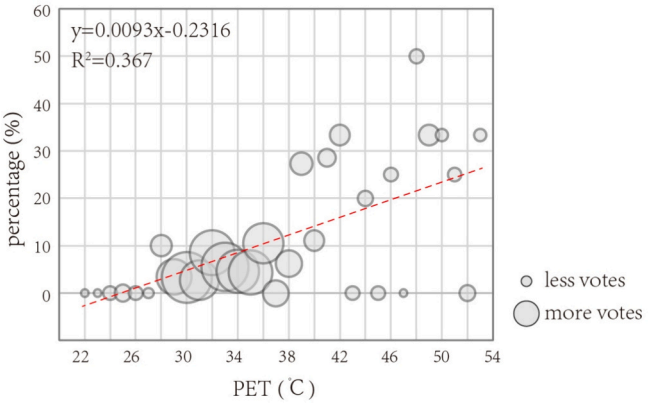


Fig. 4. Regression model of percentage of acceptable and the PET for summer

Table 6
The regression coefficient of TSV and climatic/human factors, * refers to sig.<0.05

Category	Factor	Winter	Summer
Climate factors	T_a		*
	RH		*
	V_a	*	*
	T_g	*	*
Human factors	Gender		
	Age	*	
	Length of residence		
	15m in AC places		
	Activity type	*	
	Length of stay		
	Visiting purpose		*
	Accompanied		
	Scenery		
	Overall satisfaction		

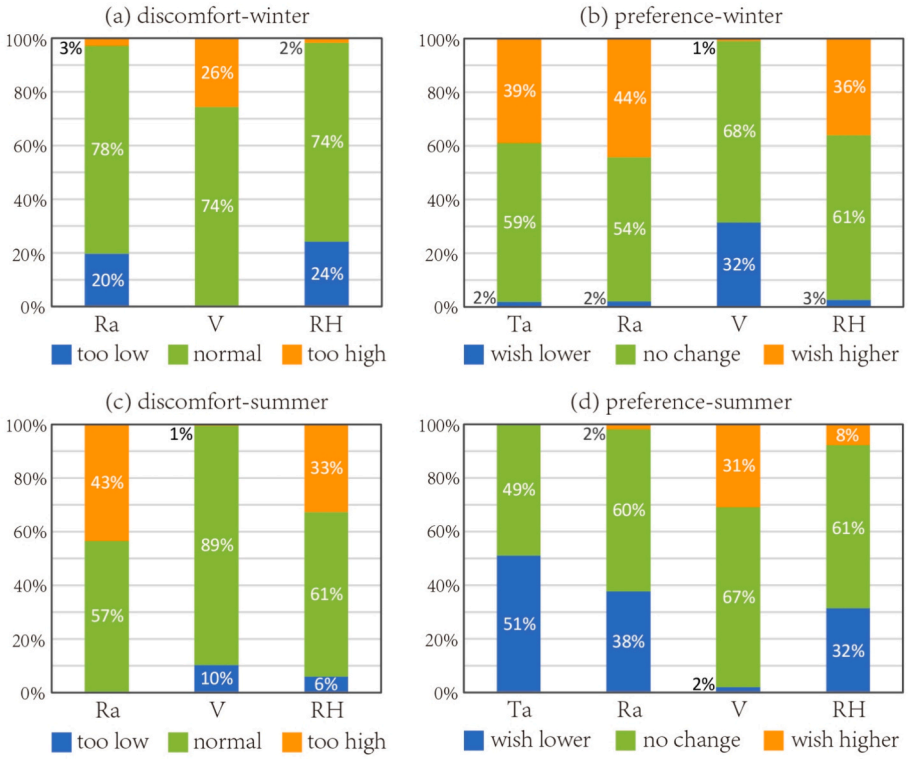


Fig. 5. Percentage of thermal discomfort and preference vote during winter and summer

and RH, respectively, to be higher. In contrast, 32% of respondents wished the wind speed to be higher. During summer (Fig. 5d), 51%, 38%, and 32% of the respondents wished T_a, sunlight, and RH, respectively, to be lower. In contrast, 31% expected a higher wind speed. It can be seen that high temperature and strong sunlight had substantial negative influence on OTC in summer. In contrast to data analysis above, many respondents felt uncomfortable due to low RH in winter.

3.4. Human factors and thermal perception

Ordinal and stepwise logistic regressions were used to analyze how human physiological and psychological factors affect OTC. The result showed that 'activity type' and 'age' has significant influence on TSV for winter, and only 'visiting purpose' has significant influence on TSV for summer (Table 6). However, this does not mean that the human body will not response or adapt to weather conditions. Therefore, this study further analyzed the relationship between each human factor and thermal perception. First, the MTSV of each option for every vote on the human physiological and psychological factors was calculated. A cross-tabulation chi-square analysis was then used to determine whether there were significant differences in the MTSV between each option (Table 7). Because

few people chose Options 1 and 2 for scenery and overall satisfaction, these two options were excluded.

For winter, four factors, namely, age, activity type, visiting purpose, and overall satisfaction, showed notable TSV differences between each option. For 'age', there was no MTSV variance between '<25' and '26–55'; however, the MTSV of '>56' was significantly higher than the other two options. For 'activity type', there was no MTSV variance between 'mid-intensity' and 'high-intensity'; however, the MTSV of 'low-intensity' was significantly lower than the other two options. For 'visiting purpose', there was no MTSV variance between 'social activity' and 'optional activity', whereas the MTSV of 'necessity activity' was significantly lower than the other two options. For the 'overall satisfaction', the MTSV increased with an enhancement in satisfaction. There are substantial MTSV variances between 'average' and 'very satisfied', whereas 'satisfied' had no difference with other options.

For summer, there were significant TSV differences between each option in four votes: under AC place within 15 min, length of stay, visiting purpose, and scenery. In the vote of 'under AC place within 15 min', The MTSV of 'Yes' was significantly higher than that of 'No'. In 'length of stay', the MTSV of '<5 min' and '>60 min' was significantly higher than the other options. There was no notable variance between the other options. In 'visiting purpose, there was no MTSV variance between 'social activity' and 'optional activity'. However, the MTSV of 'necessity activity' was significantly higher than the other two options. In 'scenery', the MTSV decreased with an enhancement in satisfaction. There was a substantial MTSV variance between 'average' and 'very good', whereas 'good' had no difference with the other options.

3.5. Proportion of elderly users in public spaces

We further obtained the proportion of elderly users (age>55) in residential and office public spaces. The proportion of elderly people between the photograph observed data and questionnaire survey results was compared. Because we cannot accurately determine the age of people in the images, the proportion of elderly people is only an approximation. It can be seen from Fig. 6 that there is substantial difference on age of attendants between residential and office areas. For the questionnaire data, during summer, 42% and 13% of respondents were aged >55 years old in residential and office areas, respectively (Fig. 6a). During winter, the percentages of people aged >55 years old were 53% and 15% in residential and office areas, respectively (Fig. 6b). For the observation data, approximately 35% and 12% of people were aged in residential and office areas during summer (Fig. 6c), and approximately 42% and 12% of people were aged in residential and office areas during winter (Fig. 6d). The proportion of aged people was smaller in the observed data because many children used public spaces but were not suitable for the questionnaire survey. Either way, the proportion of elderly users in residential areas was much higher than that in office areas.

3.6. Temporal distribution of public space users

Photo recordings were used to count the number of people per hour of the day at the seven sites. There was a substantial difference

Table 7
The MTSV of each option for each vote, * refers to significant variance in this vote (Sig.<0.05)

Winter			Summer		
Question	Options	MTSV	Question	Options	MTSV
Gender	male	-0.55	Gender	male	1.04
	female	-0.55		female	1.09
Age*	<25	-0.69	Age	<25	1.20
	26-55	-0.63		26-55	1.07
	>56	-0.41		>56	1.01
Length of residence	<1 year	-0.47	Length of residence	<1 year	1.09
	>1 year	-0.56		>1 year	1.06
15m in AC places	Yes	-0.57	15m in AC places*	Yes	1.16
	No	-0.54		No	0.98
Activity type*	low-intensity	-0.78	Activity type	low-intensity	1.02
	mid-intensity	-0.52		mid-intensity	1.09
	high-intensity	-0.37		high-intensity	1.10
Length of stay	<5m	-0.55	Length of stay	<5m	1.28
	5-15min	-0.53		5-15min	0.91
	15-30min	-0.57		15-30min	0.89
	30-60min	-0.61		30-60min	0.75
	>60min	-0.51		>60min	1.07
Visiting purpose*	social	-0.49	Visiting purpose*	social	0.95
	optional	-0.57		optional	0.89
	necessity	-0.75		necessity	1.36
Accompanied	Yes	-0.47	Accompanied	Yes	1.07
	No	-0.62		No	1.06
Scenery of this place	average	-0.68	Scenery of this place*	average	1.23
	good	-0.49		good	1.02
	very good	-0.43		very good	0.94
Overall satisfaction *	normal	-0.66	Overall satisfaction	normal	1.17
	satisfied	-0.57		satisfied	1.05
	very satisfied	-0.42		very satisfied	0.95

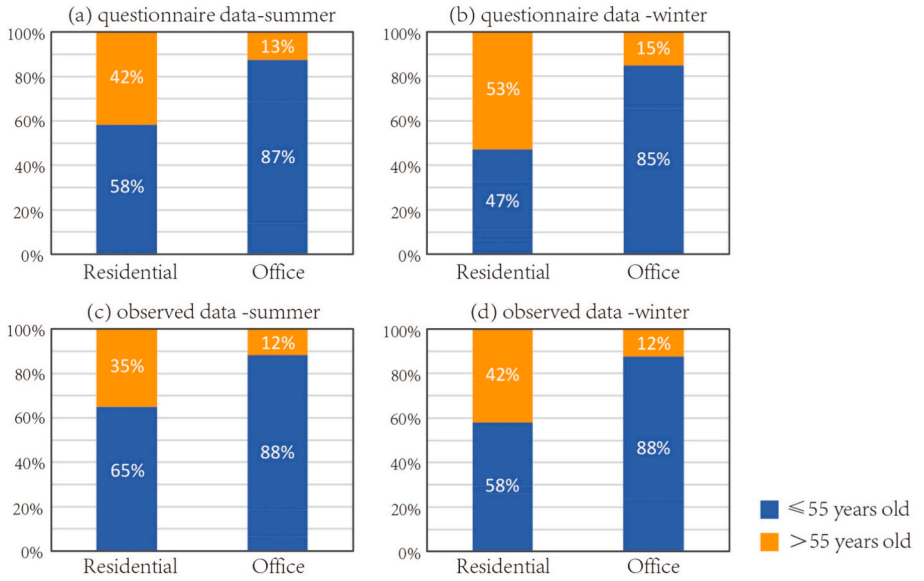


Fig. 6. Percentage of elderly people in two types of public space

in the total number of people occupying each site due to variations in the area, facilities, and surrounding population. Therefore, we converted the number of people to the percentage of people per hour to the total day attendance to compare the temporal distribution of people per day in different public spaces and seasons.

Fig. 7 shows the percentage of total day attendance per hour for the three types of public spaces during summer. There is a substantial difference in the activity time of people in each type of public space. In public spaces near residential areas (Fig. 7a), the most used hours were 09:00–11:00 and after 17:00, whereas the least used hours were between 12:00 and 15:00. There was no observable difference in the temporal distribution of attendance between weekdays and weekends in residential areas. This may be due to two reasons. First, the high air temperature and strong solar radiation at noon during summer reduce people's demand for outdoor activities. Second, most residents take lunch breaks at home in the noon. In public spaces near the office area (Fig. 7b), there was an evident peak at noon. The most used hours were between 12:00 and 13:00. This was mainly because of the after lunch break at noon for people working nearby. Public spaces near mixed areas (Fig. 7c) show a temporal distribution pattern that combines residential and office areas. On weekends, the situation of mixed areas is similar to that of residential areas. In contrast, there is an obvious peak hour at noon (12,00-13:00), which follows the pattern of office areas on weekdays.

Fig. 8 shows the percentage of total day attendance per hour for the three types of public space during winter. The temporal distribution of users in residential public spaces during winter varied greatly of that in summer (Fig. 8a). The peak hours are 10:00 to 12:00 and 14:00 to 16:00. Compared to summer, the users of residential areas during winter tended to visit near noon because of the higher temperature and stronger solar radiation. In public spaces near office areas (Fig. 8b), the temporal distribution of users during winter was consistent with that during summer, with a clear peak hour from 12:00 to 13:00. Consistent with summer conditions, the activity time pattern in mixed areas was similar to that in residential areas on weekends and showed a strong peak at noon on weekdays

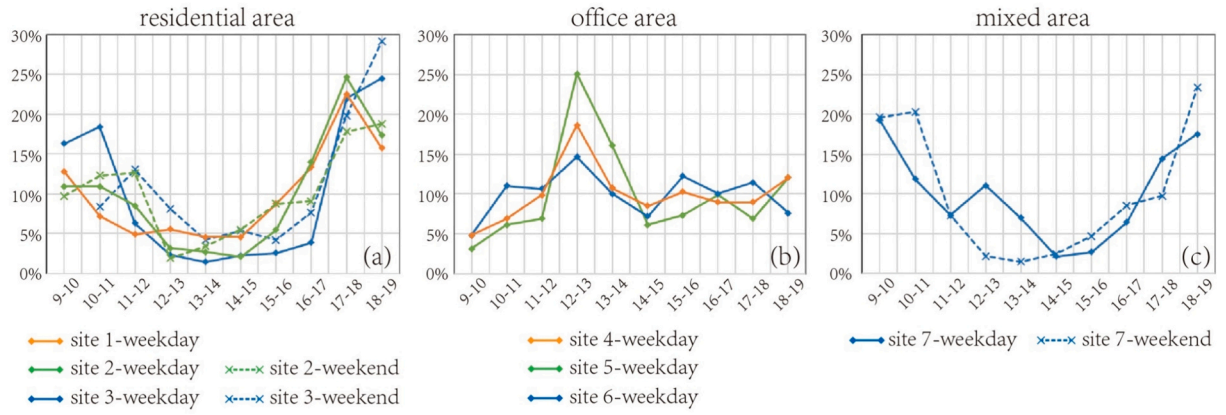


Fig. 7. Temporal distribution of public space users during summer

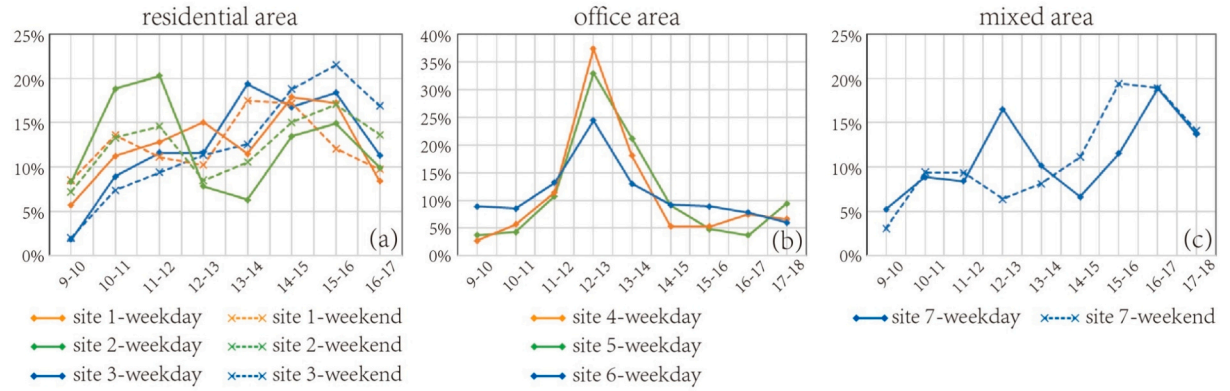


Fig. 8. Temporal distribution of public space users during winter

(Fig. 8c).

4. Discussion

4.1. NPET Comparison with other regions

This study found that the NPET of Beijing area was 21.9°C for summer and 14.3°C for winter. The NPETR was 15.2–28.6°C for summer and 6.1–22.5°C for winter. By comparing similar studies in different climate regions (Table 8), it can be found that the NPET in hotter climate regions are higher than Beijing, such as Taichung and Tehran, and the NPET in colder climate regions are lower than Beijing, such as Harbin, both for summer and winter. This indicates that local residents have adaptability to the local climate. Moreover, based on the comparison, it can be concluded that there are huge NPET differences between summer and winter (ΔPET) in areas with distinct seasons, such as Beijing (ΔPET = 7.6°C), Harbin (ΔPET = 7.7°C), Xi'an (ΔPET = 6.4°C), and Tehran (ΔPET = 7.9°C). However, for areas with little seasonal temperature variance, such as Taichung (ΔPET = 1.9°C), Melbourne (ΔPET = 4.1°C), Beer Sheva (ΔPET = 2.2°C), and Rome (ΔPET = 2.0°C), the NPET differences between summer and winter are relatively small. Therefore, we can infer that people can change seasonal OTC expectations based on their living experiences in different climates. For areas with distinct seasons, such as Beijing and other cities with comparable climatic conditions, it is necessary to develop improvement methods for outdoor thermal comfort based on different seasons.

4.2. Summer cooling or winter insulation?

This study revealed that meteorological factors have a major influence on TSV in Beijing's winter and summer. In both seasons, T_g was most influential on TSV. Therefore, changes in sunlight conditions remain essential to improving OTC. During summer, the importance of the meteorological parameters was ranked as T_g>RH>T_a>V_a. Meanwhile during winter, the essential factors that affect OTC are T_g and V_a, whereas T_a and RH are not influential. This result is consisted to existing studies (Oliveira and Andrade, 2007; Xie et al., 2022; Liu et al., 2022). However, in regions with hot summers and cold winters, there is always a contradiction between providing shade in summer and increasing sunlight in winter, as well as between summer ventilation and winter wind protection. Consequently, in Beijing, which is more important: winter insulation or summer cooling?

TSV and TCV analysis reveals that the residents of Beijing are generally less tolerant of summer heat than winter cold, suggesting a higher priority for summer cooling strategies in urban public spaces. As global warming and the urban heat island effect have raised temperatures in metropolitan areas (Tian et al., 2023), the demand for summer cooling will continue to be higher than the demand for winter insulation in Beijing or similar climate regions，due to global warming is irreversible in the near future. Additionally, people in Beijing are less tolerant to high air velocity in winter than low air velocity in summer. Although the wind speed had never exceeded

Table 8
NPET of different cities

City	Climate zone	Summer NPET (°C)	Winter NPET (°C)
Beijing (present study)	Dwa	21.9	14.3
Harbin (Yin et al., 2021)	Dwa	21.5	13.8
Xi'an (Yuan et al., 2023)	Cwa	22.3	15.9
Taichung (Lin, 2009)	Cwa	25.6	23.7
Melbourne (Kenawy and Elkadi, 2018)	Cfb	21.4	17.3
Beer Sheva (Cohen et al., 2019)	Bwh	22.8	20.6
Tehran (Hadianpour et al., 2018)	Bsk	25.1	17.2
Rome (Salata et al., 2016)	Csa	26.9	24.9

2.1m/s during winter's survey, 26% of respondents believed that the wind speed is too high. Meanwhile in summer, only 10% of respondents said low air velocity caused discomfort. Moreover, V_a is the only influential meteorological factor that affect OTC during winter in addition to T_g. This confirms the results of previous studies that wind is more influential in cold climates (Lai et al., 2020a). Thus, although summer cooling takes precedence over winter insulation, winter wind protection is more essential than summer ventilation in Beijing's public space design.

4.3. Human factors that affect OTC

Compared to meteorological factors, human physiological and psychological factors are less influential. This study only obtained three indicators that substantially impacted TSV, namely visiting purpose for summer, and activity intensity and age for winter. This result is consistent with previous studies, that human parameters are less influential under extreme climate than moderate climate (Yuan et al., 2023). The correlation between visiting purpose and TSV is as follows. People have poorer OTC in necessary activities, where the autonomy of activity is low, compared to spontaneous and social activities, where the autonomy of activity is high. However, the causal relationship between activity autonomy and thermal perception remains controversial (Lai et al., 2020a). We inferred that the extremely hot and cold climates caused a reduction in the frequency of spontaneous and social activities. Activity intensity, is positively correlated with TSV. Higher-intensity outdoor activities such as fast walking or exercise help improve OTC during winter. Public space attendants tended to have higher activity intensity during winter (avg. votes = 1.96) than summer (avg. votes = 1.72), which also proved thermal adaptation. During both seasons, the TSV of the elderly were closer to a neutral value than those of the young. This may be because the elderly is less sensitive to the weather owing to their relatively slow metabolism (Baquero and Forcada, 2022). The impact of age on OTC leads to differences in thermal comfort optimization for different types of public spaces, which is discussed in the next section.

Although the environmental stimuli factors did not exhibit a notable correlation with TSV, people tended to have better thermal comfort in public spaces with better scenery. This result is consistent with those of previous studies (Lam et al., 2020; Yan et al., 2023; Du et al., 2023). However, the causal relationship between them still requires further verification, namely, whether it is the positive environmental stimuli that enhance thermal comfort or vice versa. In order to obtain this causality, 'scenery of this place' votes in summer was calculated at an interval of 1°C PET bin, and the regression model were calculated and shown in Fig. 9. The result indicated that PET was not correlated with summer scenery (R^2=0.049, sig.=0.224). Based on this, we inferred that indeed, better environmental stimuli lead to improved thermal comfort in Beijing's summer climate. However, this issue merits more in-depth research. In addition, during summer, people who just go out from air-conditioned rooms are less tolerant to outdoor heat waves, which is consistent with some existing studies (Yang et al., 2013) but contrasts with others (Ng and Cheng, 2012; Yung et al., 2019). The relationships between them also require further investigation.

4.4. OTC and user pattern in different types of urban public space

Field investigation results exhibited a substantial discrepancy in the temporal distribution of people during the day in different seasons and different types of public spaces. In residential area, people tend to choose more comfortable times to perform outdoor activities during both seasons. This demonstrates that climatic conditions have a considerable influence on people's behavior. On the contrast, for both seasons, people in office areas tend to use public spaces around midday, which proves that the work schedule has a stronger influence on people's outdoor activities than climate conditions. This phenomenon highlights the necessity of targeted OTC optimization strategies for two types of urban public spaces. In general, sunlight protection is more important for office areas than for residential areas in summer because outdoor spaces are more commonly used during the hottest time of the day. At this time, the solar elevation angle is large, making it difficult for buildings to provide effective shading to outdoor spaces. Tall trees with large canopy and higher leaf area index, can effectively provide shade, reduce T_g, and improves outdoor human thermal comfort during hot seasons (Irmak et al., 2018; Broadbent et al., 2018; Li et al., 2023). Thus, compared to residential areas, increasing the tree coverage in public spaces of office areas, can more effectively improve thermal comfort. Some local deciduous trees, such as pagoda trees and poplars, provide effective summer shade without blocking winter sunlight. Moreover, although this study's survey ended at 19:00, it can be inferred that public spaces in residential areas may still gather a significant number of users after sunset during summer. Therefore, providing adequate lighting is crucial for enhancing the utilization efficiency of public spaces in residential areas.

In addition to differences in time periods, the proportion of elderly individuals in residential areas is significantly higher than in office areas. As analyzed in the previous section, elderly individuals are less sensitive to high summer temperatures and low winter temperatures compared to younger individuals. Although the mTSV of elderly citizens were closer to neutral, lower climate sensitivity may lead to health risks when elderly individuals are exposed to extremely hot or cold weather (Larriva and Higueras, 2020). As there were more older attendees, residential areas need to pay more attention to extreme weather conditions than office areas. According to Larriva and Higueras's (2020) study, activity facilities are important factors in encouraging elderly individuals to use public spaces. Therefore, compared to office areas, activity facilities in residential areas, such as fitness equipment, chess tables, and benches, should be more rigorously placed in locations with favorable microclimates.

Previous studies have demonstrated that urban geometry is the major ways to change outdoor microclimates (Jamei et al., 2016; Lai et al., 2019). Under Beijing's climatic conditions, summer shade and cooling are more important than winter insulation, whereas the need for winter wind protection is stronger than summer ventilation. Therefore, a compact urban layout is suggested for the Beijing area to reduce the sky view factor and increase the building coverage ratio. A lower sky view factor contributes to summer shade and cooling (Zhang et al., 2020; Kim and Brown, 2022), whereas a higher building coverage ratio is beneficial for reducing wind speed (Ma

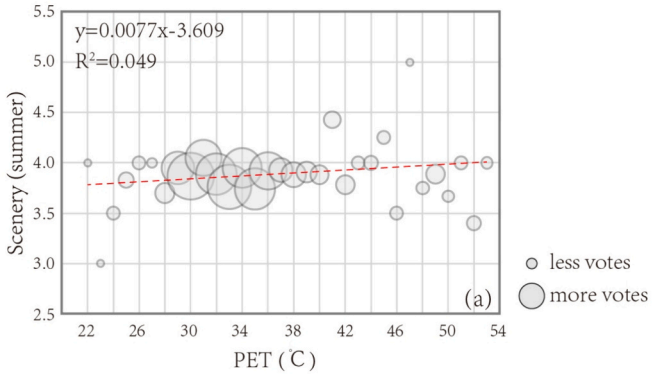


Fig. 9. Regression model of scenery and PET for summer

and Chen, 2020). Despite this, we also suggest creating a concentrated open space in a compact urban layout to receive sufficient sunlight during winter.

4.5. Limitation of the study

The selection of respondents in this study has limitations. Firstly, in urban public spaces, the majority of people did not engage in high-intensity activities, resulting in a smaller sample size for this group. Secondly, there were fewer child respondents because very young children might not fully comprehend the questionnaire. Thirdly, since the survey period coincided with the ongoing COVID-19 pandemic, the number of tourists was very low, which might have affected the impact of length of stay on OTC. Finally, the survey was conducted during the hottest and coldest periods of the year, causing some individuals sensitive to extreme weather to intentionally reduce their outdoor time. Therefore, future research needs to select other locations and methods to conduct more targeted studies on the influence of certain human characteristics on OTC.

5. Conclusion

Using a questionnaire survey, on-site microclimate measurements, and photo documentation, this study investigated the outdoor thermal comfort in several public spaces in Beijing during summer and winter. The following conclusions were drawn. The founding of this study provides theoretical support for the optimization of outdoor thermal comfort in Beijing and regions with similar climates.

1) There was a considerable OTC discrepancy between winter and summer under Beijing's local climate conditions. The NPET is 21.9°C for summer and 14.3°C for winter, whereas the NPETR is 15.2–28.6°C for summer and 6.1–22.5°C for winter.

2) Meteorological parameters have a dominant effect on the OTC in Beijing's climate. During summer, the sequence of influential weights was T_g>RH>T_a>V_a. During winter, the sequence of influential weights on OTC was T_g>V_a, whereas RH and Ta showed no notable impacts. This seasonal variation highlights the necessity of targeted OTC optimization for different seasons in a cold winter and hot summer region.

3) In Beijing's outdoor spaces, summer cooling is more important than winter insulation, whereas winter wind protection is more essential than summer ventilation.

4) Human physiological and psychological parameters, such as age, activity intensity, and visiting purpose, have certain effects on OTC. However, they are much less influential than meteorological parameters. Short-term thermal experiences and environmental stimuli also have a minor impact on thermal comfort, but the relationship between them requires further study.

5) There is considerable discrepancy in the user pattern in different types of public spaces and seasons. In residential areas, people choose suitable times for outdoor activities based on the weather. However, in office areas, the timing of people's activities is dictated by work schedules. This discrepancy has a significant influence on the OTC optimization in different public spaces.

6) For Beijing's local climatic conditions, a compact urban layout with concentrated open spaces is recommended to obstruct sunlight during summer and cold winds during winter. Planting local deciduous trees is an effective way to improve OTC, especially for public spaces in office areas, where people are exposed to extreme solar radiation at noon during summer.

Funding information

This research is supported by NSFC project: The mechanism and optimization of thermal comfort for urban public space in cold region based on human thermal sensation difference (Grant No. 52208002), and NSFC project: Theory, method and mechanism of monitoring and early warning for the implementation of spatial planning based on evidence-based practice (Grant No. 52078008).

CRediT authorship contribution statement

Tong Ma: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jian Zhang:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. questionnaire design

Outdoor thermal comfort questionnaire

questionnaire No._____time_____location_____weather: sunny/partly cloudy/cloudy

interviewee under direct sunlight: ①Yes ②No

Q1: Gender ①male ②female

Q2: Age group ①≤15 ② 16-25 ③ 26-40 ④ 41-55 ⑤56-65 ⑥ > 65

Q3: You are ①living nearby ②working nearby ③others

Q4: How long have you been stay in Beijing?

①<3 months ②3-6 months ③6-12 months ④more than a year

Q5: in past 15 min, have you been stay in indoor/cars with air-conditioning?

①Yes ②No

Q6: in past 15 min, what is your main body activities?

①sitting ②standing ③walking slowly ④walking fast ⑤light exercise ⑥heavy exercise ⑦take care of children ⑧others

how long have you been stay there?

①<5min ②5-15min ③15-30min ④30-60min ⑤>1hour

Q8: what is your purpose to be in this place?

①rest/recreation/sightseeing ②doing exercise (alone) ③doing exercise (group) ④social activities/meeting friends ⑤just walking by ⑥waiting for someone ⑦Taking care of children ⑧others

Q9: are you here alone, or with others? ①alone ②with others

Q10: How do you think the scenery of this place?

①very bad ②bad ③average ④good ⑤very good

Q11: Do you overall satisfied with this place?

①very unsatisfied ②unsatisfied ③normal ④satisfied ⑤very satisfied

Q12: How do you feel in terms of thermal perception now?

①very cold(1) ②cold(2) ③slightly cold(3) ④neutral(4) ⑤slightly hot(5) ⑥hot(6) ⑦very hot(7)

Q13：Beside air temperature, which following factors make you uncomfortable at this time? (multi-choice)

①wind: too strong/too weak ②sunlight: too strong/too weak ③humidity: too strong/too weak

Q14: What would you say the overall thermal comfort at this place?

①comfortable ②slight uncomfortable but acceptable ③uncomfortable

Q15: Which of the following factors do you think would make your more comfortable? (multi-choice)

①temperature: lower/higher ②sunlight: lower/higher ③wind: lower/higher ④humidity: lower/higher

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